BACHELOR'S THESIS DEFENSE

Training Neural Networks with Evolution Strategies

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Bachelor's Degree in Artificial Intelligence

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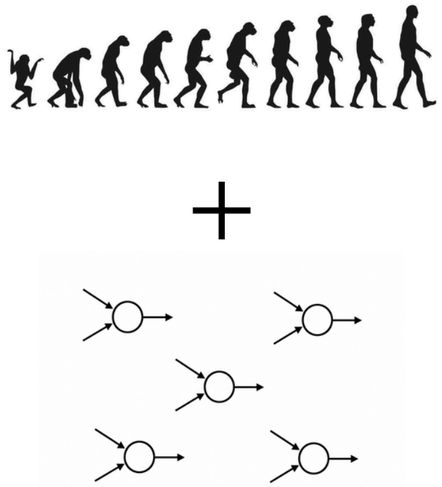
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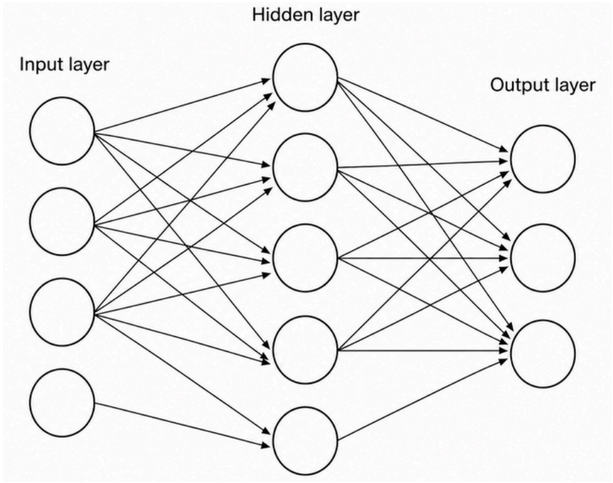
CORE IDEA

A different evolutionary representation

Instead of evolving directly a complete neural network, this thesis evolves the neurons that build one network using Evolution Strategies.



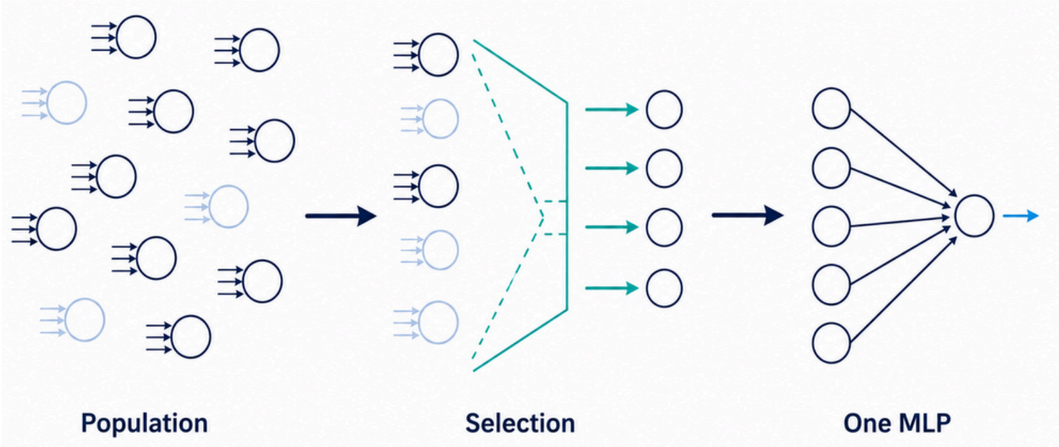




CORE IDEA

From evolved neurons to one Multilayer Perceptron

Candidate neurons are selected; the survivors become the hidden layer of one final network.



MOTIVATION

Why this thesis?

A different way to train neural networks

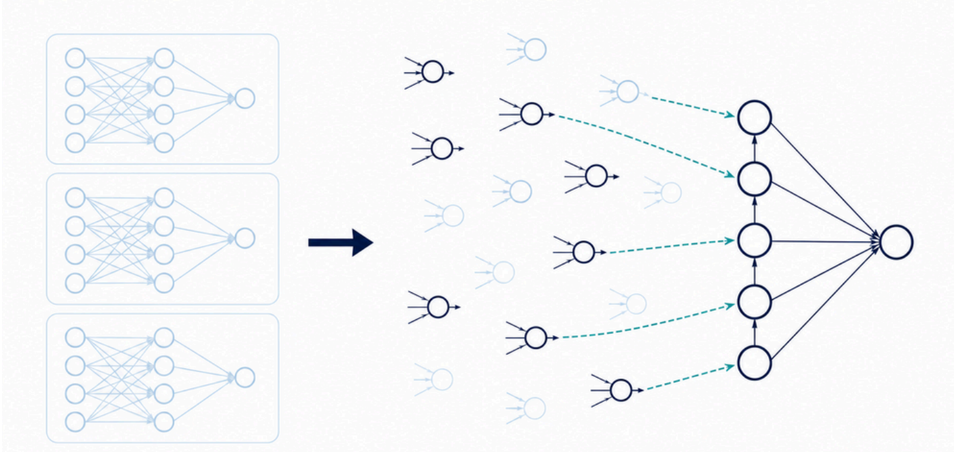
✗ No backpropagation through the hidden layer

Representation changes the problem

Population can become one cooperative hidden layer

A possible accuracy–cost trade-off

No need of repeatedly evaluating many high-dimensional complete networks



OBJECTIVES

Research steps

- [ ] Implement a working Michigan-style ES-MLP prototype.
- [ ] Design a coherent neuron-level fitness strategy.
- [ ] Evaluate binary, regression and multiclass variants.
- [ ] Compare the representation against Pittsburgh-style evolution.

SCOPE

Scope of the thesis

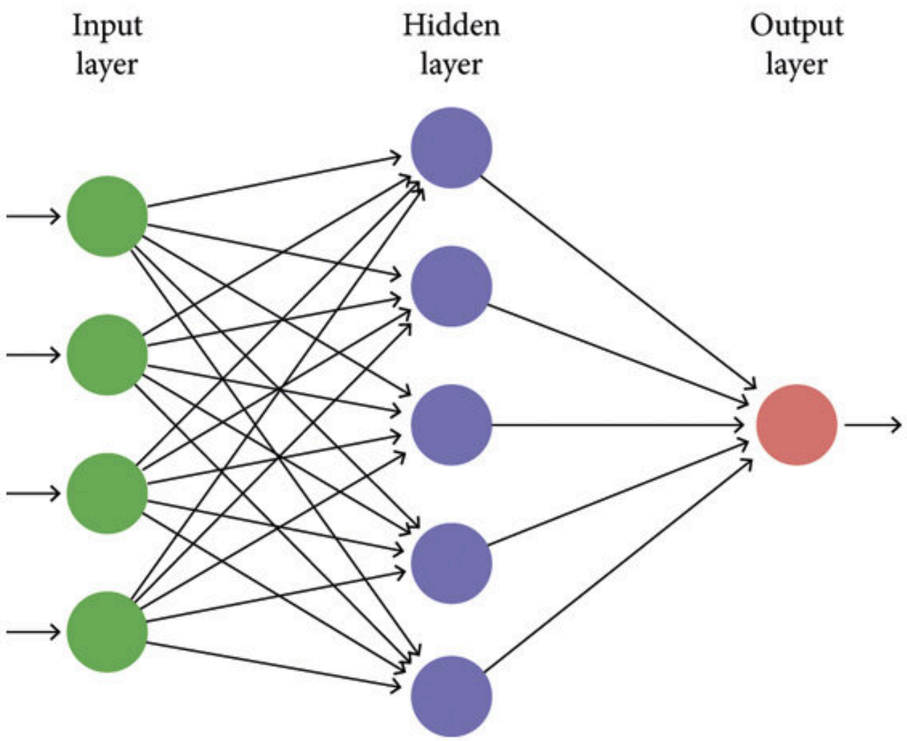
It studies

- A compact single-hidden-layer MLP for tabular data.
- Michigan-style Evolution Strategies at neuron level.
- Task-specific output layers.

It does not claim

- To replace backpropagation in deep learning.
- To be universally better than classical tabular models
- To evaluate every neuroevolution variant.

Single-hidden-layer MLP



$$h_i(x) = \tanh(w_i^T x + b_i)$$

A hidden neuron transforms the original tabular input into one nonlinear activation.
The hidden layer becomes a learned feature map.

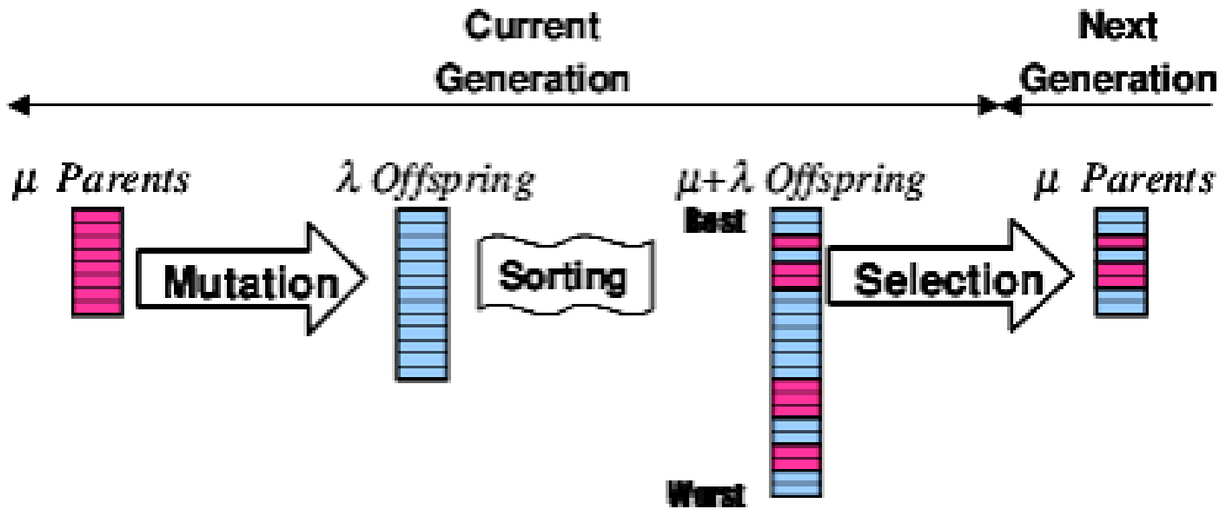
Why not just use backpropagation?

Backpropagation

Updates all trainable weights through gradients of a loss function.

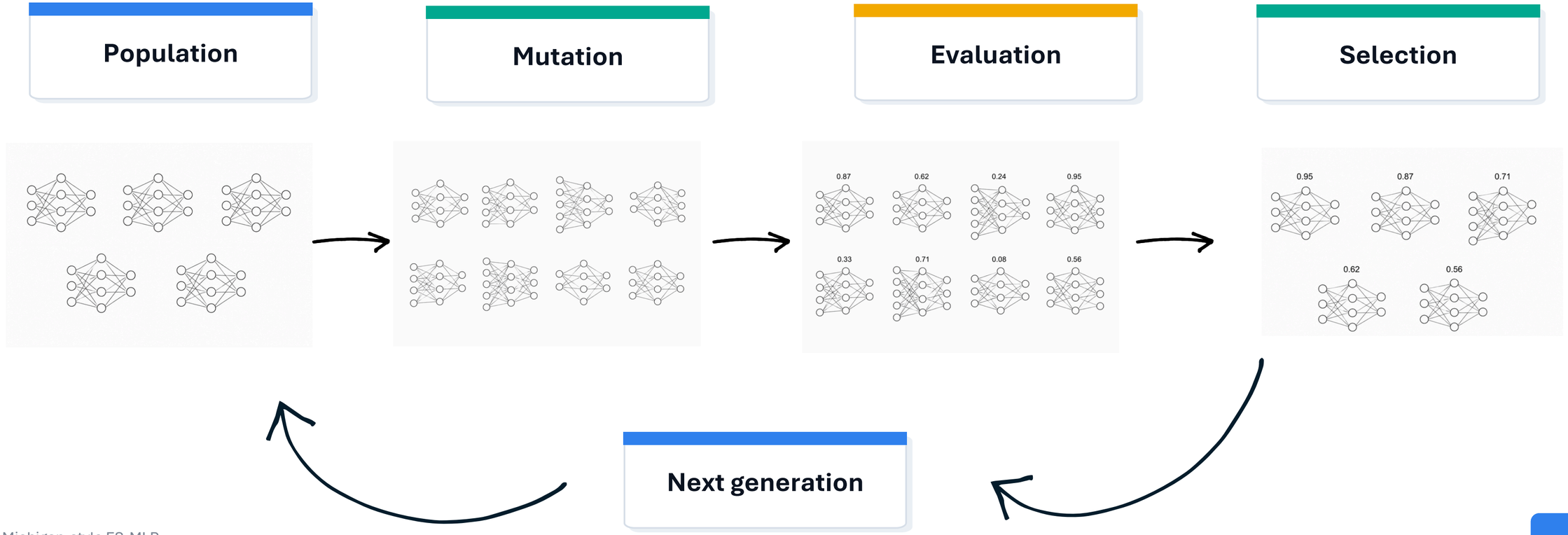
Evolution Strategy

Generates candidate solutions by mutation and keeps candidates according to selection.

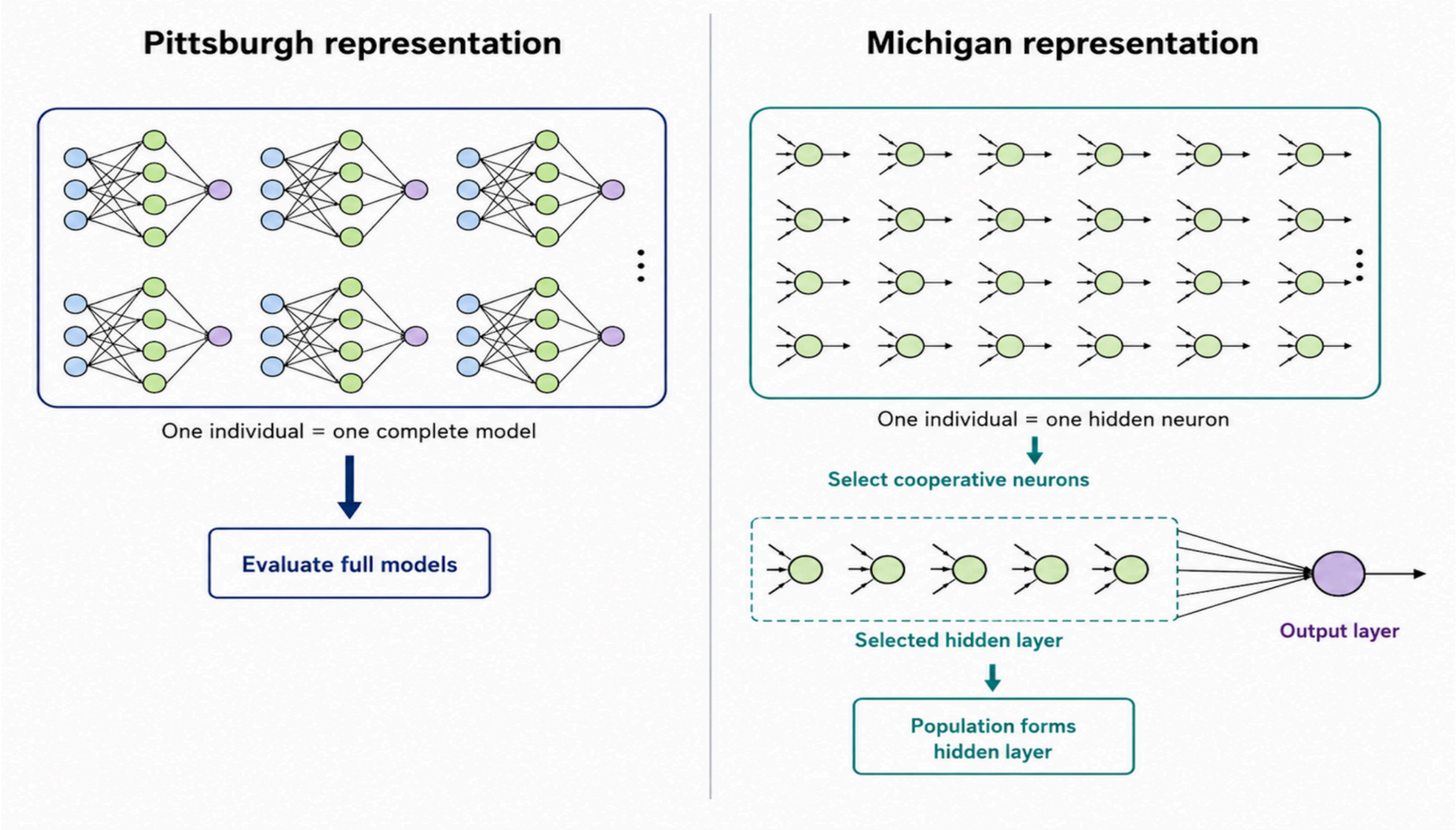


Neuroevolution: what is usually evolved?

The most common case is to evolve a full network, where one individual is already a complete model.

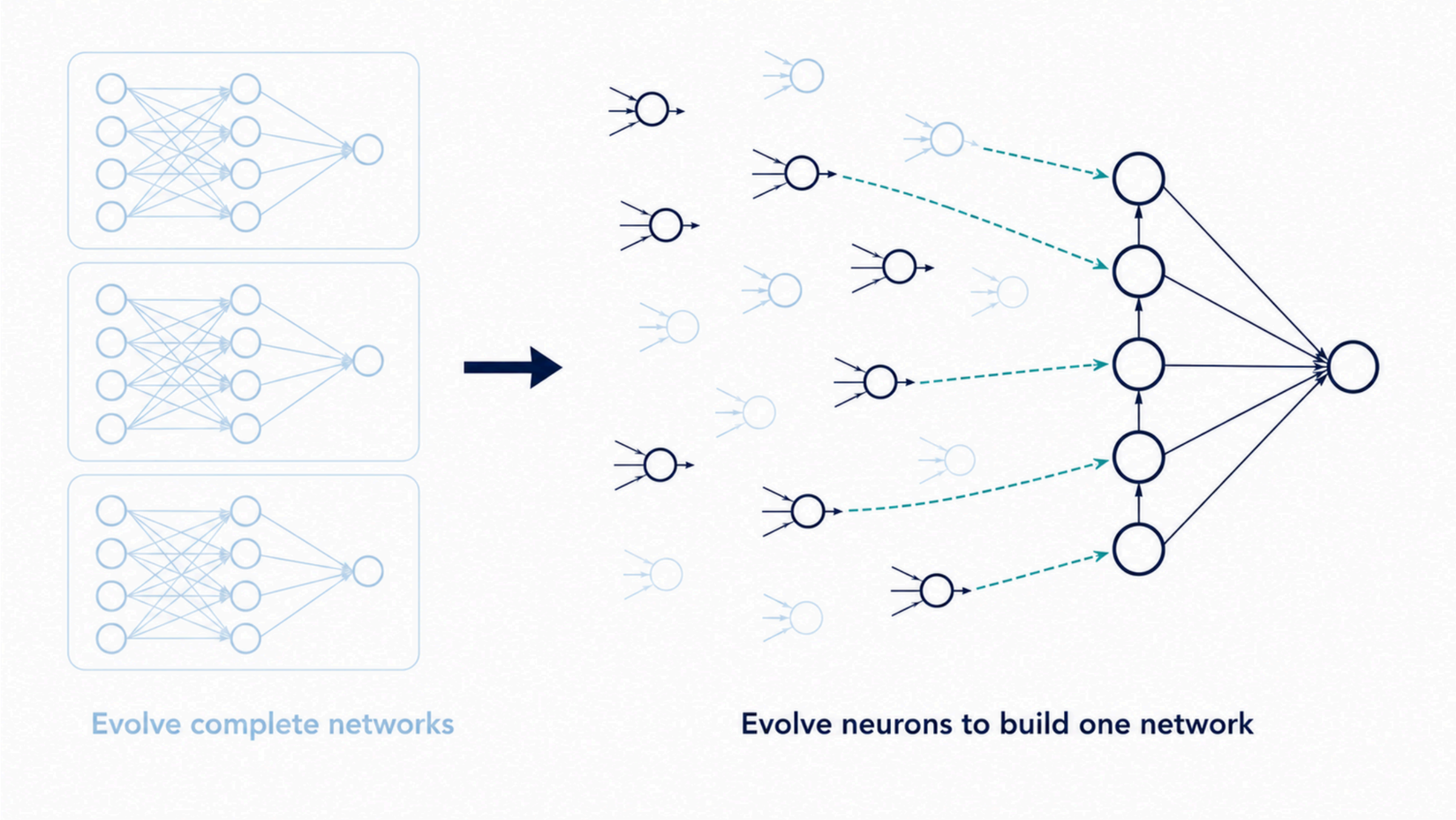


Pittsburgh vs Michigan



REPRESENTATION

Pittsburgh vs Michigan



REPRESENTATION

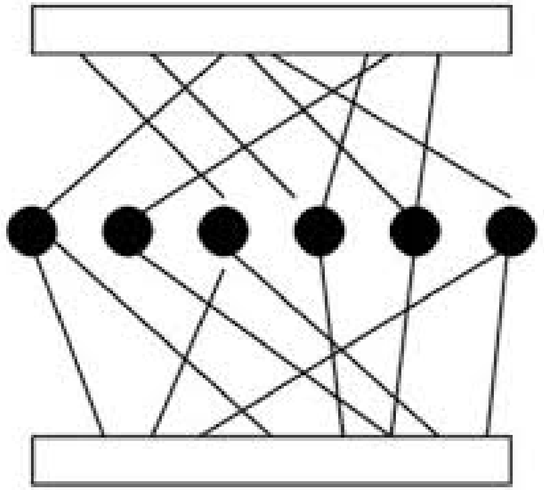
Pittsburgh vs Michigan

	Pittsburgh	Michigan
Individual	One neural network	One hidden neuron
Population	A set of competing candidate networks	A set of cooperating neurons forming one hidden layer
Fitness	Direct model performance	Cooperative neuron selection
Final model	Best neural network	Keep the selected neuron population as the final hidden layer

Cooperative neuroevolution

- SANE (1996) and CoSyNE (2008)
- A good component is one that improves the network together with others.

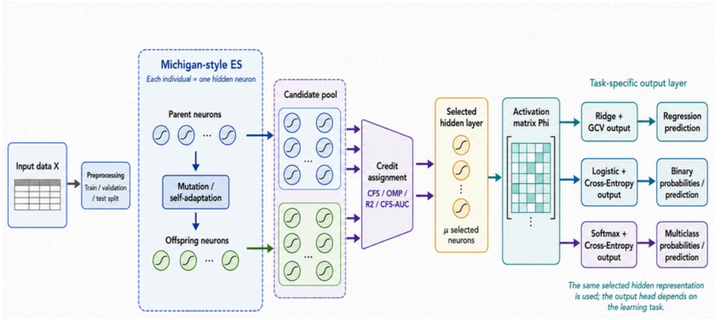
A neuron may be weak alone but valuable because it adds information that the other neurons do not provide.



RESEARCH GAP

Positioning of this thesis

A Michigan-style Evolution Strategy for a single-hidden-layer MLP



- Neurons are selected as nonlinear features: relevant to the target and complementary to each other
- The output layer is fitted separately for the task

Neuron selection → cooperative nonlinear feature selection

Prototype overview

Individual

One hidden neuron

Population

The hidden layer

Parent

A neuron that survived the previous generation.

Offspring

A new candidate neuron obtained by mutating a parent neuron.

Candidate pool

Parent neurons + offspring

Fitness strategy

Criterion used to decide which candidate neurons should survive.

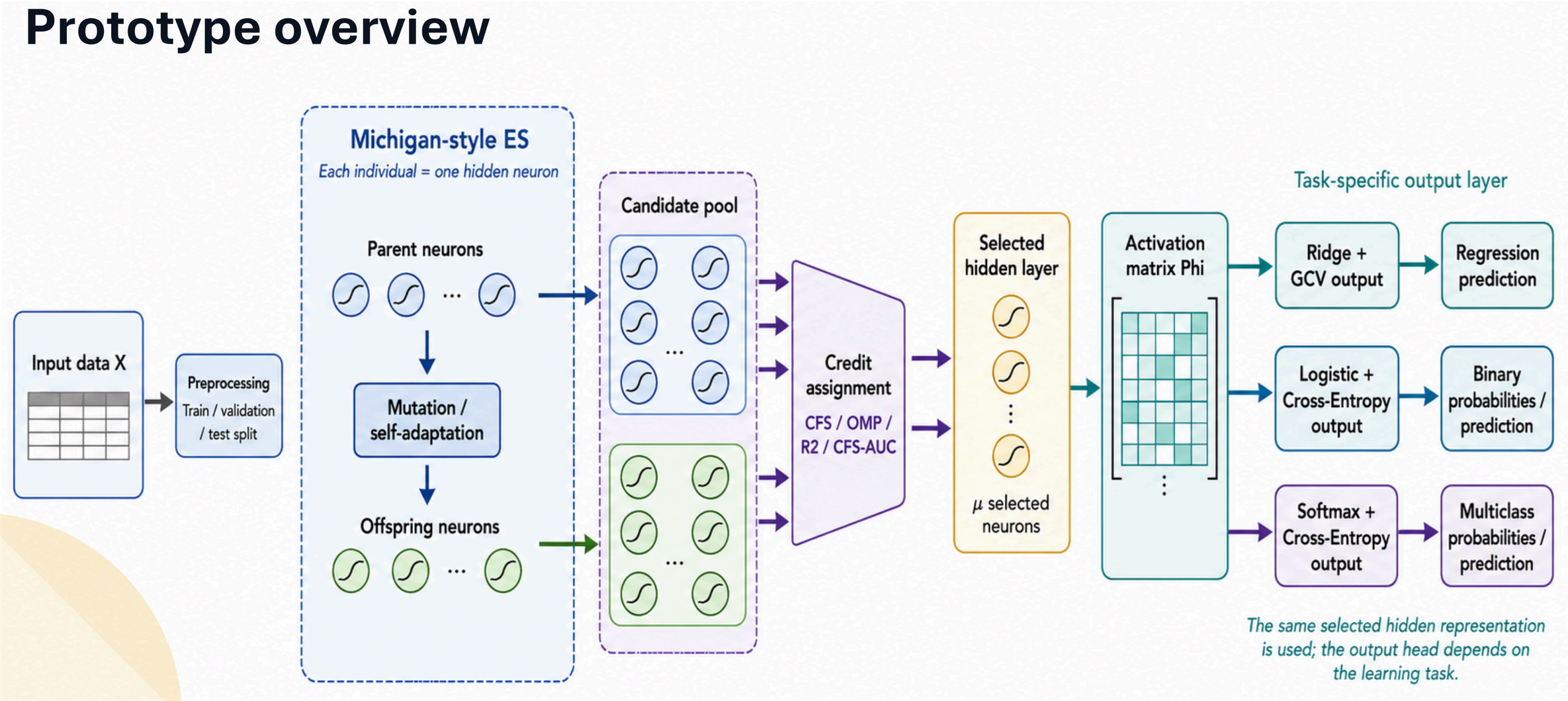
Selection

The operation that keeps μ neurons from the candidate pool.

Output fitting

The supervised step applied after hidden-neuron selection

Prototype overview



Each individual is one hidden neuron

Each neuron has its own mutation strategy, allowing different local mutation scales and directions during evolution.

$$\boldsymbol{\alpha}_i = [w_{i,1}, \ldots, w_{i,n}, b_i]^\top \in \mathbb{R}^D, \qquad D = n + 1.$$

α: a matrix of neuron genotypes with shape μ × D

L: mutation-strategy matrix for neuron i, with shape μ × D × D

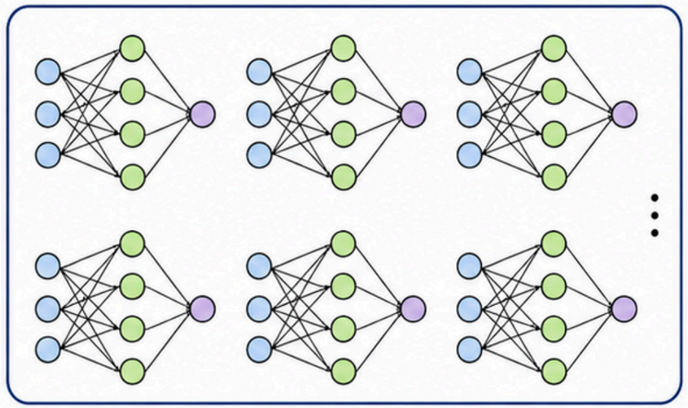
Mutation covariance $\longrightarrow$ $\boldsymbol{\Sigma}_i = \mathbf{L}_i \mathbf{L}_i^\top$

KEY CHALLENGE

Fitness strategy: easy for a full network, hard for one neuron

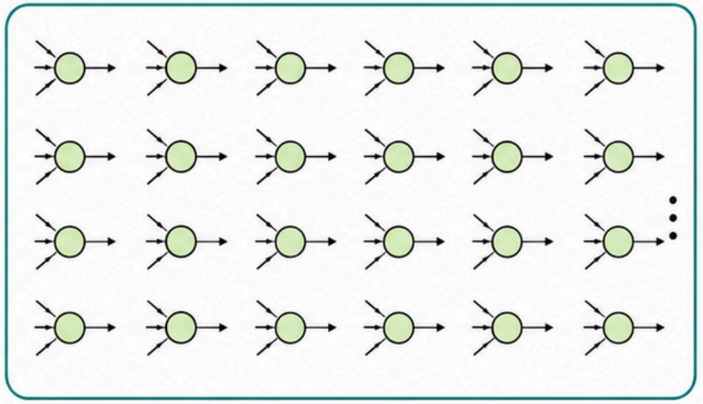
Pittsburgh

One individual is a complete model.
Accuracy or loss can directly be its fitness.

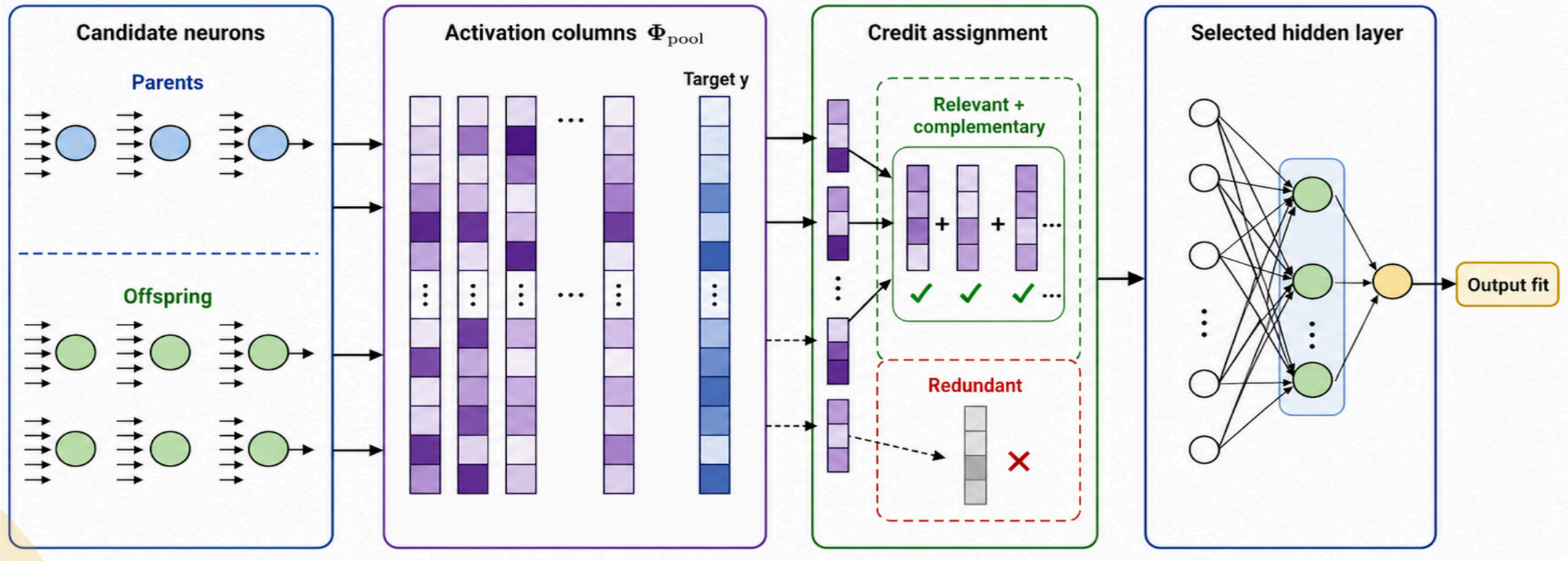


Michigan

One individual is only one neuron.
How can we decide which neurons are the ones that should survive?



Neuron selection as nonlinear feature selection



FITNESS STRATEGIES

Four selection criteria tested

Marginal R^2

Rewards individual neuron-target correlation with a diversity penalty.

OMP

Selects neurons according to residual information.

CFS

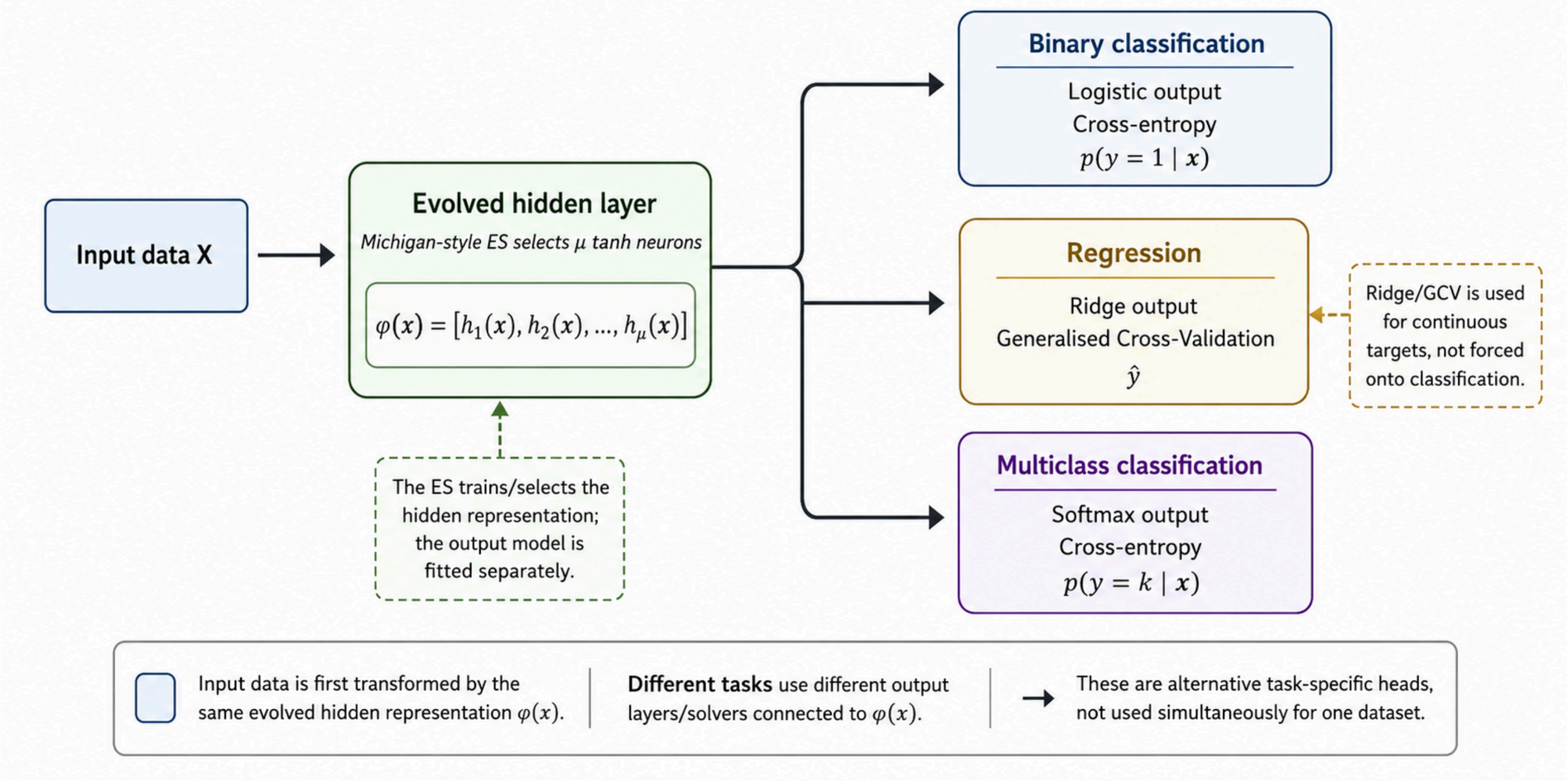
Rewards target relevance and penalises redundancy.

CFS-AUC

Uses AUC relevance and Spearman redundancy.

OUTPUT LAYER

Same hidden representation, task-specific output



How results are evaluated

Train

Preprocessing

ES evolves the hidden representation

Validation

Select checkpoint, strategies and hyperparameters

Test

Final evaluation only

Never used for model selection

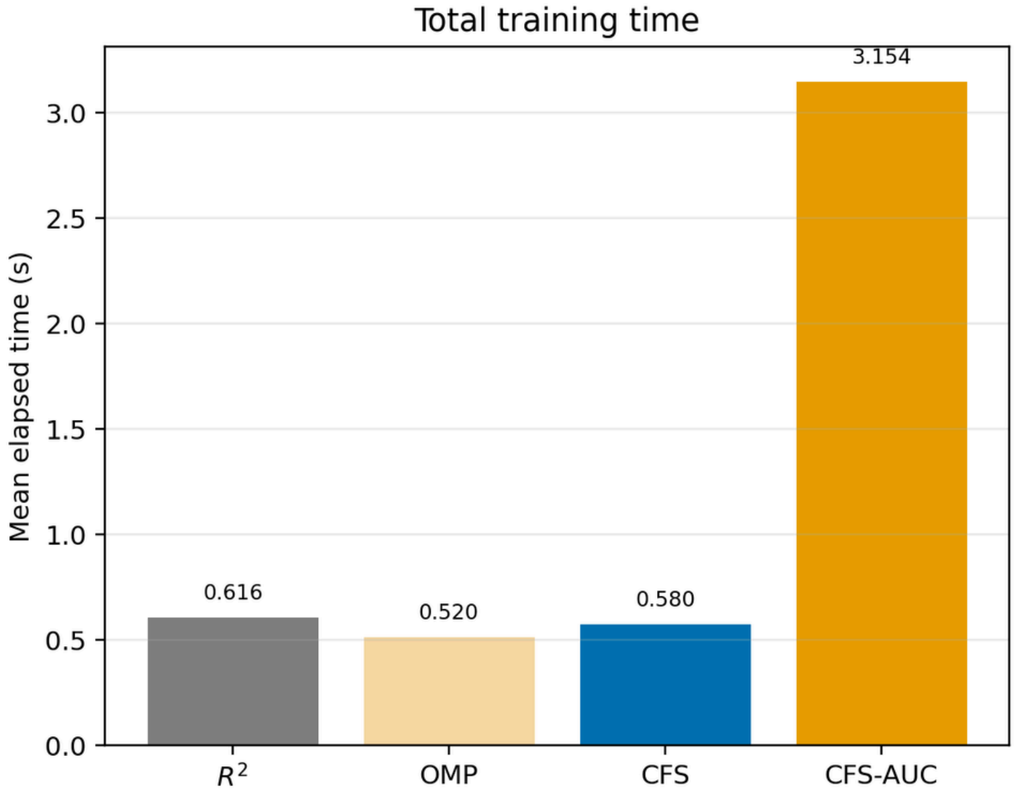
What does the experimental work assess?

- Can the method identify neurons that are useful together?
- Does the evolved hidden representation add value over simple alternatives?
- Can the same representation support different task-specific output heads?
- Can the hidden layer be reduced while preserving performance?
- Does neuron-level evolution offer a favourable accuracy–time trade-off against Pittsburgh-style evolution?

RESULT

Neuron-selection strategy comparison

Strategy	Val Acc	Test Acc	Test CE
OMP	0.901 ± 0.070	0.868 ± 0.084	$\mathbf{1.428} \pm 0.827$
R^2	0.909 ± 0.075	0.872 ± 0.092	2.724 ± 2.678
CFS	$\mathbf{0.920} \pm 0.060$	$\mathbf{0.881} \pm 0.072$	2.117 ± 1.286
CFS-AUC	0.917 ± 0.050	0.859 ± 0.087	2.651 ± 1.592



DEFAULT STRATEGY

Why CFS became the main choice

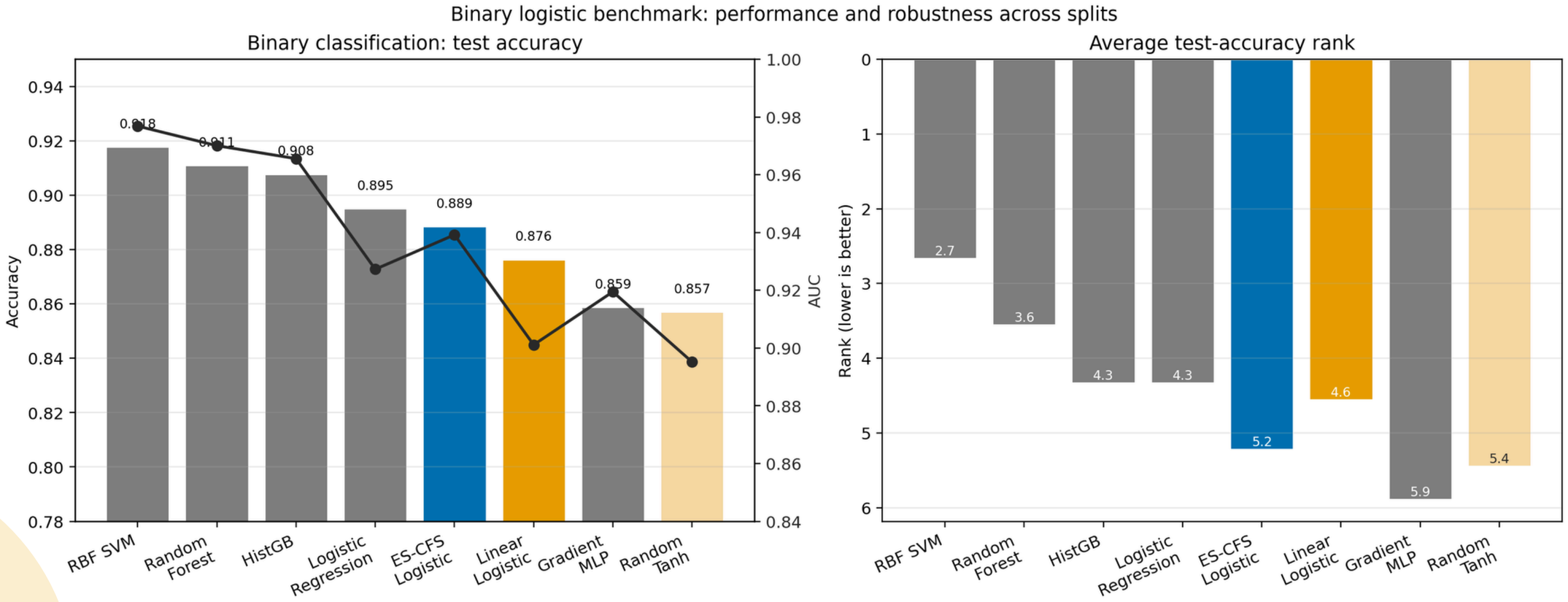
A good hidden layer needs neurons that are individually relevant, but also complementary.

Relevance to the target + low redundancy between selected neurons

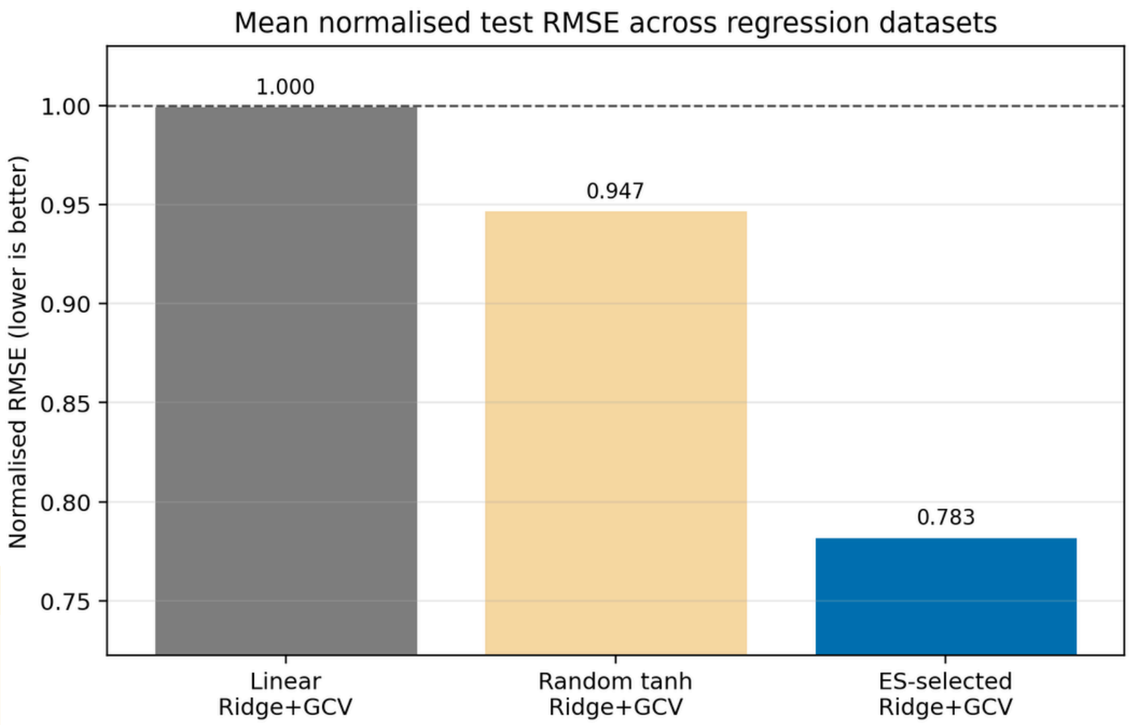
For a subset S of size k, the merit is:

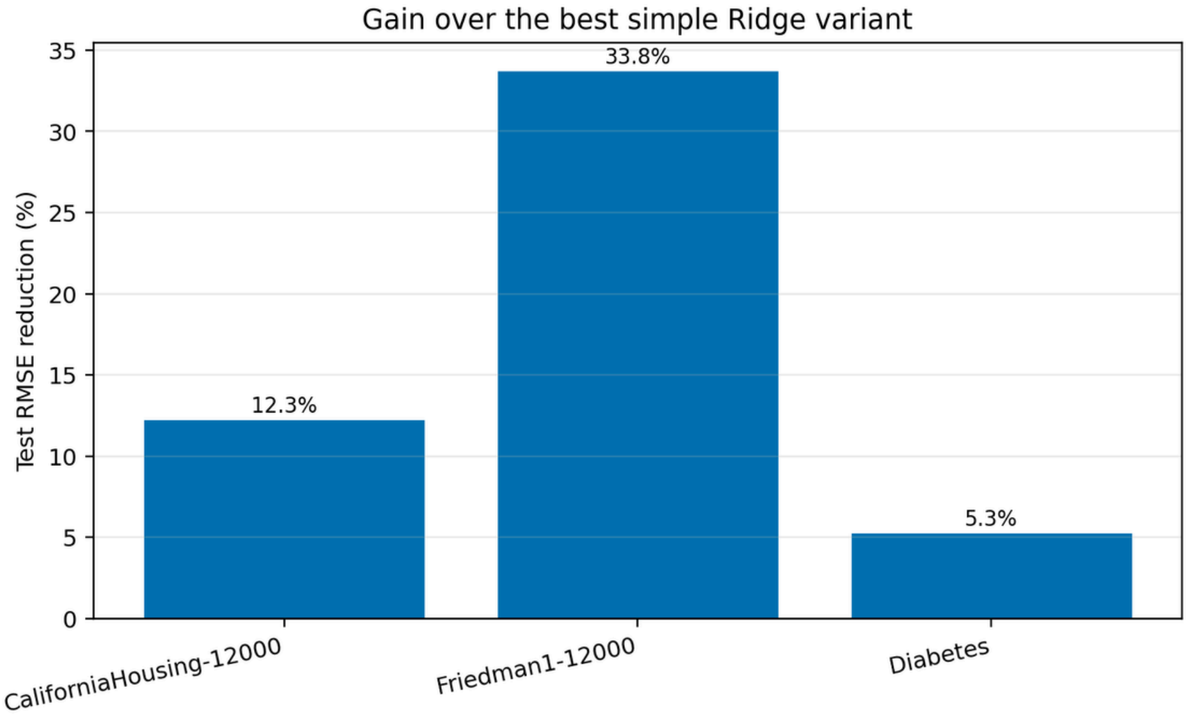
$$\mathrm{Merit}(\mathcal{S}) = \frac{k\bar{r}_{cf}}{\sqrt{k + k(k-1)\bar{r}_{ff}}},$$

ES-CFS Logistic CE vs tabular baselines

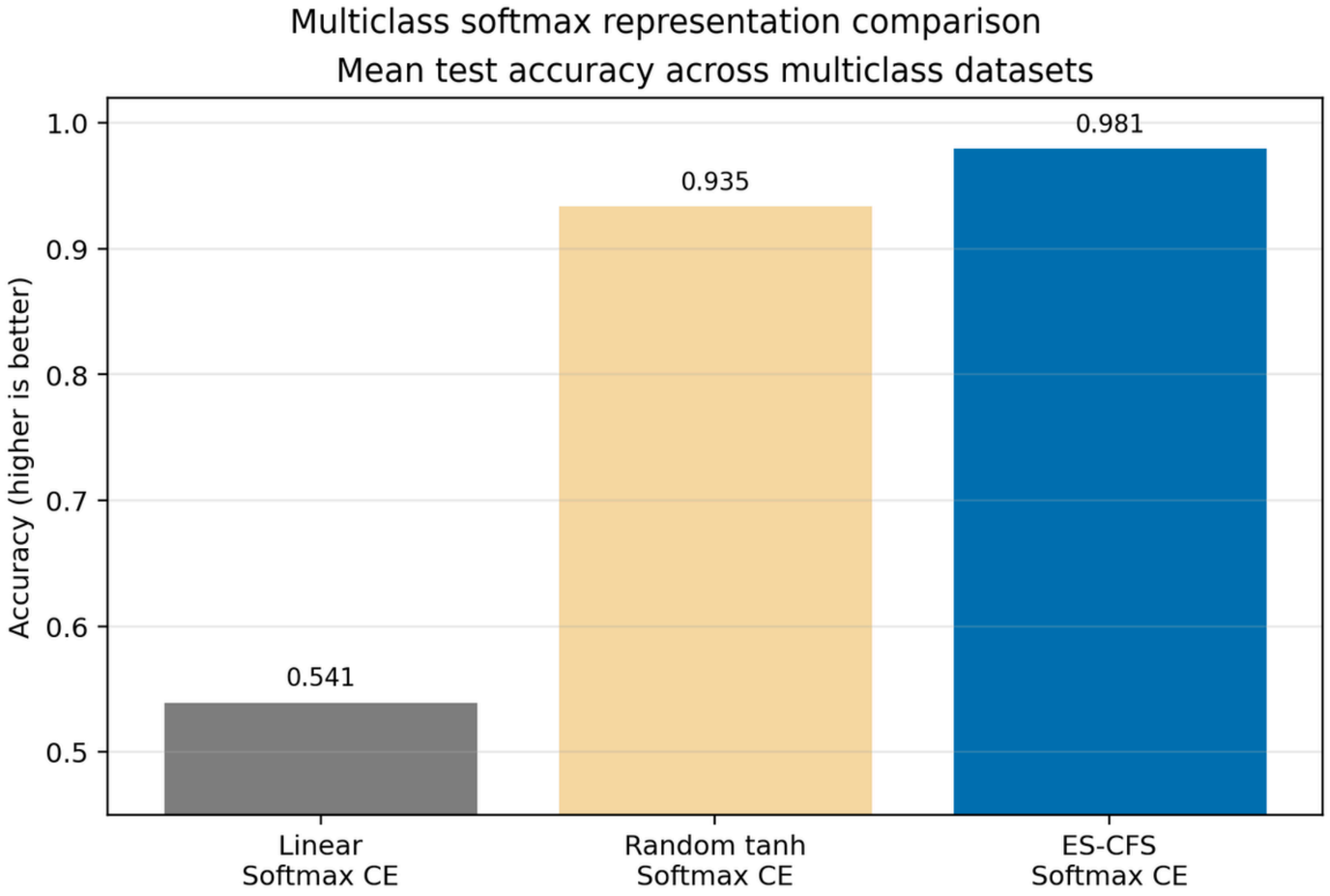


Ridge/GCV for regression tasks





Softmax head

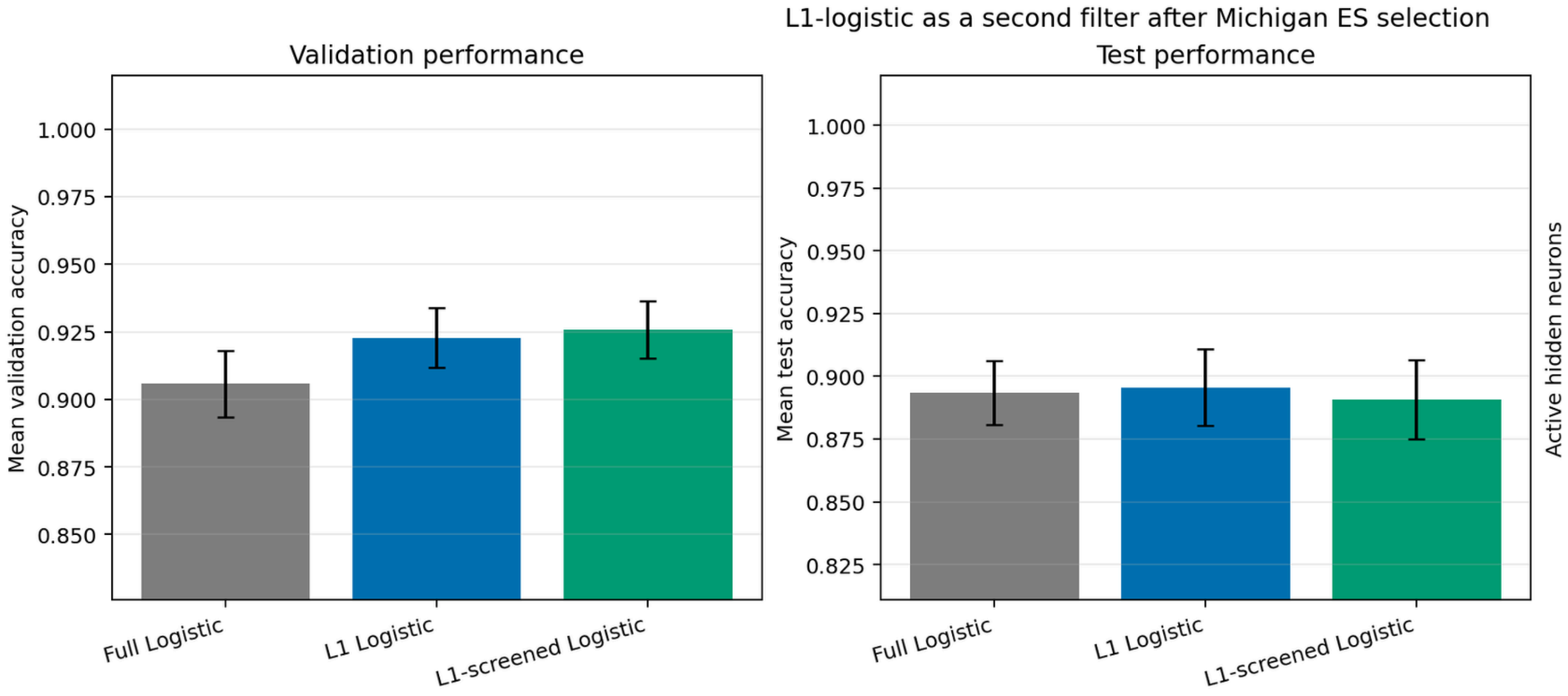


SPARSITY

L1-logistic as a second filter

After ES selects 50 neurons, L1-logistic can remove neurons by setting output coefficients to zero.

Similar performance with fewer active neurons



Direct L1 retains regularisation in the final predictor.
Screened L1 uses L1 only for pruning.

Similar performance with fewer active neurons

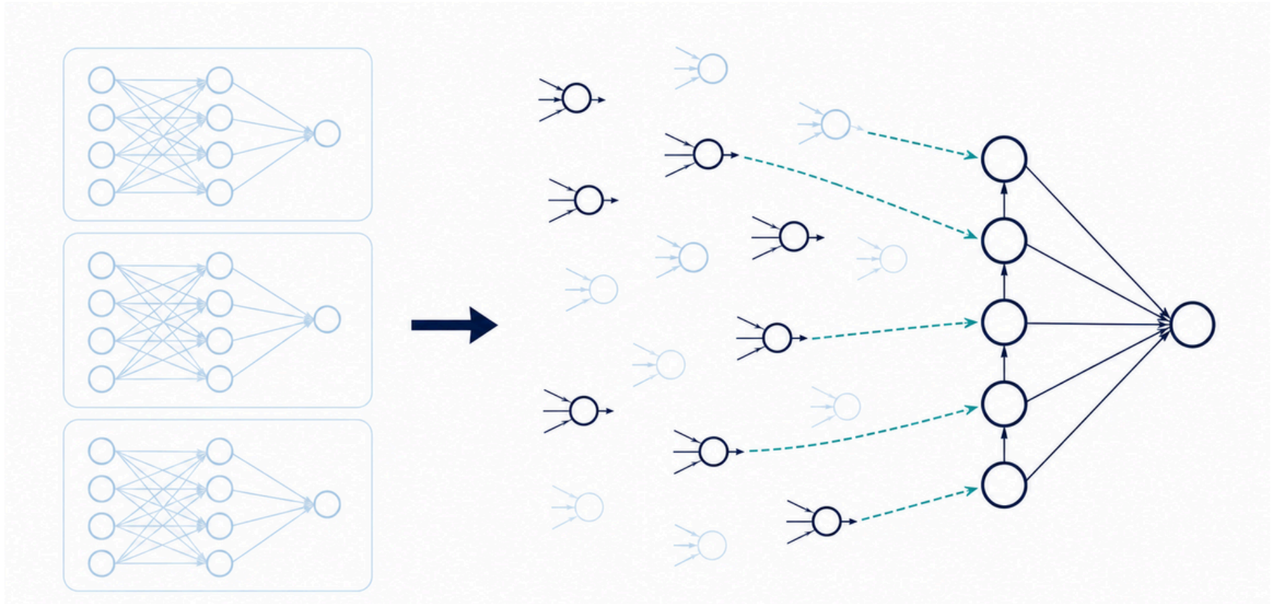
Table 11: L1-logistic filtering experiment over Breast-W, Ionosphere, and Sonar with ten seeds. Values are reported as mean ± standard error across seeds.

Output layer	Val Acc	Test Acc	Test AUC	Active neurons	Active fraction
Full ES-CFS Logistic CE	0.906 ± 0.012	0.893 ± 0.013	0.939 ± 0.009	50.00 ± 0.00	1.000 ± 0.000
ES-CFS L1 Logistic	0.923 ± 0.011	$\mathbf{0.896} \pm 0.015$	$\mathbf{0.950} \pm 0.010$	18.47 ± 1.64	0.369 ± 0.033
L1-screened ES-CFS Logistic CE	$\mathbf{0.926} \pm 0.011$	0.891 ± 0.016	0.942 ± 0.011	$\mathbf{17.53} \pm 1.37$	$\mathbf{0.351} \pm 0.027$

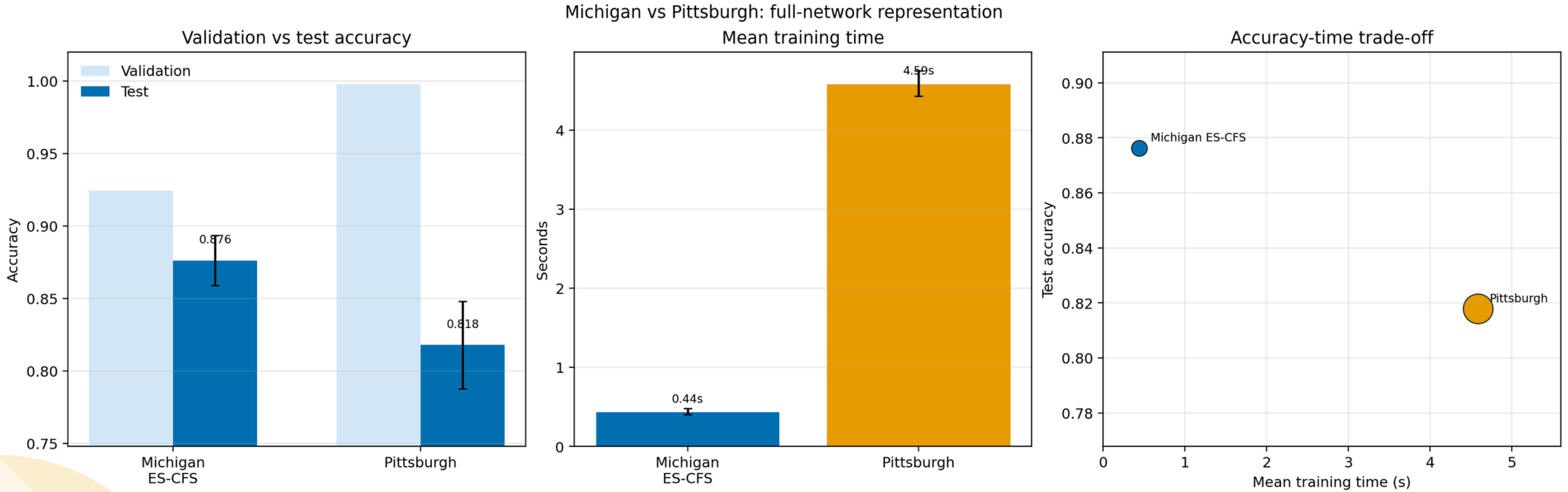
FINAL EXPERIMENT

Michigan vs Pittsburgh

This experiment directly tests whether evolving individual neurons can provide a better accuracy–time trade-off than evolving complete neural networks.

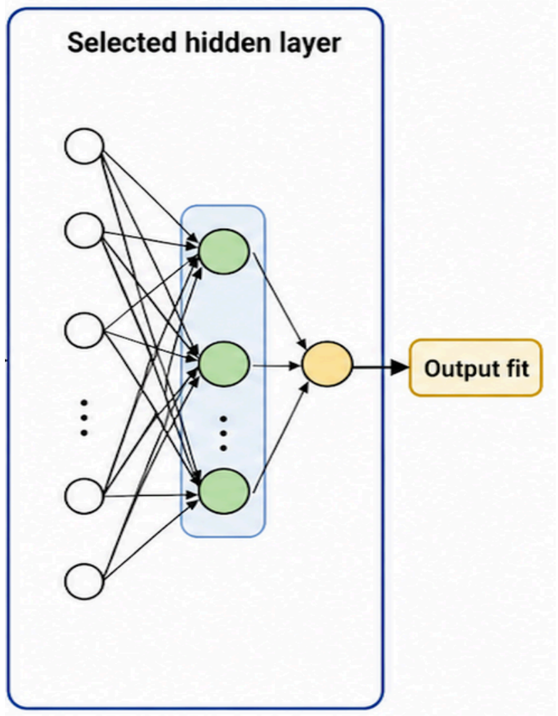


REPRESENTATION RESULT



What the thesis supports

- Neuron-level Michigan ES is a viable way to build a hidden representation.
- The fitness strategy is central; CFS gives the best overall balance.
- Task-specific output layers avoid mixing incompatible objectives.
- The approach is competitive, but not universally superior.



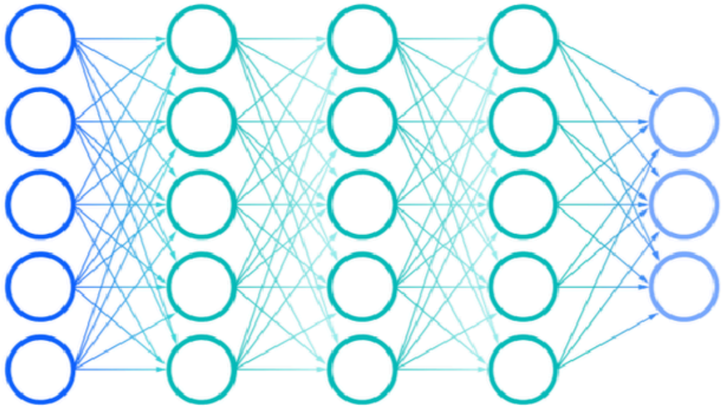
OBJECTIVES

Research steps

- [x] Implement a working Michigan-style ES-MLP prototype.
- [x] Design a coherent neuron-level fitness strategy.
- [x] Evaluate binary, regression and multiclass variants.
- [x] Compare the representation against Pittsburgh-style evolution.

Future work

- The prototype is focused on one hidden layer and fixed hidden width.
- A deeper version would require credit assignment across layers.
- Letting the number of hidden neurons vary would make the architecture more flexible.
- Both extensions are large enough to motivate future work.



Thank you for your attention

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